

SANRAKSHA SOS SAFETY APP

ABSTRACT

Sanraksha SOS Safety App is an Android-based personal safety application developed to support users during emergency situations. The main objective of the project is to provide a fast, reliable, and simple way to request help when the user feels unsafe.

Many people, especially women, students, elderly individuals, and solo travelers, face situations where immediate communication is critical. In such cases, opening multiple apps or making a manual phone call may take too much time.

Sanraksha addresses this problem by offering one-touch and automatic emergency alert mechanisms. The application is built using Kotlin for Android and is designed with practical real-world usability in mind.

It allows the user to activate a Safety Mode before travel or while entering a risky environment. Once enabled, the app continuously supports emergency monitoring in the background. The system can trigger SOS manually, through voice keywords such as help commands, or through shake-based phone motion detection.

This reduces response time in situations where the user cannot unlock the phone or type a message. When SOS is activated, the app sends alert messages to saved emergency contacts using SMS. The alert message includes user details and location information whenever available.

A countdown-based cancellation feature is also provided to reduce false alerts. The project further supports location tracking and periodic journey updates through a connected cloud database. This helps trusted contacts monitor the user's movement during travel.

The app includes emergency contact management, profile storage, siren control, and bilingual support in English and Hindi. Its design focuses on accessibility, speed, and low user effort during stress conditions. The project demonstrates how mobile sensors, GPS, SMS, speech recognition, and background services can be combined for public safety.

The outcome of the project is a functional safety solution that improves emergency communication and increases the chances of timely assistance.

Sanraksha is therefore a socially relevant and technically practical final year engineering project with strong real-world application.

INTRODUCTION

Personal safety has become a major concern in modern society due to increasing cases of harassment, assault, accidents, and unsafe travel conditions. Women, children, elderly people, students, and people who travel alone are especially vulnerable in emergency situations. In many such cases, immediate communication with trusted persons can make a significant difference. However, during panic situations, users may not have enough time to unlock their phone, dial a number, or type a message. This creates a strong need for smart and fast emergency response systems.

Emergency response systems are important because they help reduce the time between danger detection and help notification. A well-designed system can instantly alert family members, friends, or authorities with essential information such as the user's location. Quick access to such information improves the chance of timely rescue and support. Mobile-based safety applications are especially useful because smartphones are widely available and equipped with sensors, GPS, internet, microphone, and messaging features. Despite the availability of safety apps, many existing solutions have limitations. Some applications depend only on internet connectivity, which makes them unreliable in low-network areas.

Others require too many manual steps, which is not practical during fear, confusion, or physical restriction. Some apps do not support automatic trigger methods such as voice commands or motion detection. In addition, many solutions do not provide multilingual accessibility or continuous background support during travel.

Sanraksha SOS Safety App is developed to address these practical challenges through a simple and reliable Android-based solution. The app allows users to activate a Safety Mode and trigger SOS through multiple methods such as manual activation, voice keywords, and shake detection.

It sends emergency SMS alerts to saved contacts along with the user's location whenever available. The application also supports journey monitoring through background location updates and cloud storage. Additional features such as emergency contact management, profile details, siren control, and language options improve usability.

Thus, Sanraksha provides a practical, fast, and user-friendly approach to personal safety and emergency communication.

PROBLEM STATEMENT

In emergency situations, the biggest challenge is the delay in communicating with people who can provide immediate help. Traditional emergency response methods such as phone calls or manual texting often require time, attention, and physical control of the device.

During panic, injury, assault, or sudden danger, a user may not be able to unlock the phone or type a message quickly. This delay can reduce the chances of timely support and increase the level of risk.

Therefore, there is a need for a faster and more accessible emergency communication system. Many existing safety solutions are not fully reliable in real-world conditions. Several applications depend mainly on internet connectivity to send alerts or share live location.

In areas with poor network access, such systems may fail to deliver the emergency message at the required time.

This creates a serious problem for users travelling through rural areas, highways, or low-signal zones. A dependable safety system must support offline alert mechanisms in addition to online services. Another problem is that many current applications require the user to perform multiple manual steps before help can be requested.

This is not suitable in situations where the user is under stress, under attack, or unable to interact normally with the device. Lack of automatic triggering methods makes these systems less practical in real emergencies. Users need features such as voice-based activation, shake detection, and one-touch SOS support for faster response.

Real-world users also face issues such as travelling alone at night, feeling unsafe in public transport, sudden medical distress, or being unable to explain their location clearly. In such moments, trusted contacts need immediate information including the user's identity and current location. Most existing methods do not combine quick alerting, location sharing, offline messaging, and background monitoring in a simple interface.

There is also limited support for personalized safety information and multilingual accessibility in many available solutions. As a result, users remain dependent on slow, incomplete, or unreliable emergency communication methods.

The Sanraksha project is proposed to solve this problem by providing a fast, practical, and mobile-based SOS safety system.

OBJECTIVES

- To design and develop an Android-based personal safety application for emergency situations.
- To provide instant SOS alert generation for faster communication with trusted contacts.
- To send emergency messages through SMS so that alerts can work even without internet access.
- To include location details in alert messages for quick identification of the user's position.
- To enable one-touch manual SOS activation during panic situations.
- To support hands-free emergency triggering using voice keyword recognition.
- To support automatic SOS triggering through shake detection when manual access is difficult.
- To reduce communication delay during unsafe, stressful, or medically critical situations.
- To provide a Safety Mode that monitors the user during travel or vulnerable conditions.
- To allow users to store and manage emergency contact details within the application.
- To maintain user profile information such as name, gender, blood group, and medical details.
- To improve the chances of receiving timely help through fast and structured alert messages.
- To support real-time or periodic location tracking for journey monitoring.
- To integrate cloud-based location updates for improved remote visibility of user movement.
- To provide a cancellation timer to avoid false or accidental SOS alerts.
- To offer multilingual support for better accessibility and ease of use.
- To create a simple, clear, and user-friendly interface suitable for real-world use.
- To increase user confidence while travelling alone or facing risky environments.
- To combine mobile sensors, GPS, SMS, and background services into one practical safety solution.
- To develop a socially relevant engineering project with real-time safety benefits and practical application.

LITERATURE REVIEW

Personal safety applications have gained importance with the growth of smartphone-based emergency support systems. Many existing safety apps are designed to help users send alerts, share live location, or contact emergency services during danger.

Common examples in this domain include women safety apps, travel safety tools, guardian-tracking systems, and emergency messaging applications. These systems generally use smartphone resources such as GPS, internet connectivity, SMS, and contact lists. Their main goal is to reduce communication delay and improve emergency assistance.

Several existing applications provide useful features such as one-tap SOS buttons, live location sharing, and emergency contact notification. Some apps also include siren alarms, route monitoring, and cloud-based safety tracking.

A few advanced systems support voice input or wearable device integration for emergency triggering. These developments show that mobile technology can play an important role in personal safety and rapid response.

However, practical usability remains a major concern in many currently available solutions. One common limitation is the strong dependence on internet connectivity for sending alerts or updating location.

In real emergency situations, internet access may be weak, unavailable, or unstable. Another limitation is that many apps require multiple manual actions before an alert is sent. This reduces effectiveness when the user is under panic, stress, or physical threat.

Some systems also lack automatic trigger mechanisms such as shake detection or continuous voice-based monitoring. Limited multilingual support and insufficient personalization further reduce accessibility for diverse users. Sanraksha improves upon these limitations by combining offline SMS alerts, location sharing, and multiple trigger methods in one Android application.

It supports manual SOS activation, voice keyword recognition, and shake-based triggering for faster emergency response. The application also includes Safety Mode, emergency contact management, user profile details, and bilingual support for better usability.

Thus, Sanraksha offers a more practical, accessible, and real-world oriented safety solution compared to many conventional mobile safety applications.

METHODOLOGY

The Sanraksha system follows an event-driven mobile application methodology for emergency detection and response. The application is developed as an Android-based system using Kotlin and integrated device services. Its working begins with user input through different emergency activation methods.

The input may be a manual SOS button press, a shake gesture, or a predefined voice command. These multiple input options are provided to ensure activation even in stressful or restricted conditions. Once an input is received, the application checks whether Safety Mode is active and SOS conditions are valid.

The trigger detection module processes the event and starts the emergency workflow. The system then fetches user profile details and emergency contact information stored in the application. After that, it attempts to obtain the current GPS location using Android location services.

If location is available, the app prepares a message with map link and coordinates. If location is unavailable, the system still sends the alert with the available emergency information. The message generation process includes the user's identity, essential medical details if provided, and emergency text.

Before final alert transmission, a short cancellation countdown is given to avoid false alarms. After confirmation or countdown completion, the output stage begins. The output includes sending SMS alerts to saved emergency contacts and showing app status notifications to the user.

At the same time, background services may continue monitoring location and safety state. The main modules of the system include the SOS trigger module, shake detection module, voice activation module, GPS tracking module, contact management module, and alert transmission module.

The Android architecture is based on activities for user interaction and foreground/background services for continuous monitoring tasks. SharedPreferences is used for local storage of user profile, language settings, and emergency contacts.

For backend support, Firebase Realtime Database is used as an optional cloud service to upload periodic journey location data for remote monitoring.

SYSTEM REQUIREMENTS

A. Hardware Requirements

1. Android smartphone with minimum basic processing capability.
2. Device with GPS module for location tracking.
3. Device with accelerometer sensor for shake detection.
4. Microphone support for voice command recognition.
5. SIM-enabled phone for sending SMS alerts.
6. Battery support sufficient for foreground safety services.
7. Internet connectivity for optional cloud synchronization.
8. Internal storage for app installation and local data saving.

B. Software Requirements

1. Android Studio as the development environment.
2. Kotlin programming language for application development.
3. Android SDK for building and testing the application.
4. Minimum Android version compatible with app deployment.
5. Google Play Services Location API for GPS access.
6. Gradle build system for dependency and project management.

C. Packages/Tools Used

1. `play-services-location` for location services.
2. `androidx.localbroadcastmanager` for in-app broadcast communication.
3. Material Design components for interface elements.
4. Firebase Realtime Database for cloud-based location upload.
5. Android `SmsManager` for SMS alert transmission.
6. Android Speech Recognition through `RecognizerIntent` for voice triggering.
7. Android Foreground Services for continuous background monitoring.
8. SharedPreferences for local storage of user and settings data.

D. Optional Requirements

1. Firebase backend for storing trip and journey location records.
2. Cloud connectivity for remote tracking through database integration.
3. Speech recognition support as a lightweight voice-based detection mechanism.
4. Future AI/ML integration can improve keyword recognition and threat prediction.
5. Optional web dashboard support can help contacts monitor travel activity.
6. Additional analytics tools can be integrated for safety pattern analysis.

- **ADVANTAGES**

1. Works even without internet by using SMS-based emergency alerts.
2. Provides faster emergency communication during critical situations.
3. Reduces delay in informing trusted contacts.
4. Offers a simple and easy-to-use mobile interface.
5. Supports multiple trigger methods such as button press, voice command, and shake detection.
6. Helps users activate SOS even when manual phone operation is difficult.
7. Shares location details to improve rescue and response accuracy.

- **DISADVANTAGES**

- 1) Continuous background services may increase battery consumption.
- 2) GPS tracking and sensor monitoring can reduce device power efficiency over long usage.
- 3) SMS alerts depend on the availability of a mobile network signal.
- 4) Emergency messages may be delayed in areas with weak cellular coverage.
- 5) Voice-based triggering may not work properly in noisy environments.
- 6) Shake detection can sometimes produce false triggers during normal phone movement.

CONCLUSION

The Sanraksha SOS Safety App is a practical mobile application developed to improve personal safety during emergency situations. The project successfully combines essential smartphone features into one integrated emergency response system. It provides users with a fast and accessible way to send alerts when they feel unsafe or face sudden danger.

The application supports manual SOS activation, voice-based triggering, and shake detection for flexible emergency use. It also sends SMS alerts with location details, which improves the chances of timely assistance. The inclusion of offline SMS support makes the system more useful in low-internet or no-internet conditions.

Safety Mode and background monitoring further increase the effectiveness of the application during travel or risky situations. The project also includes contact management, profile storage, bilingual support, and optional cloud-based journey tracking.

These features make the application user-friendly, practical, and suitable for real-world deployment. From an engineering perspective, the project demonstrates the effective use of Android services, sensors, GPS, SMS, and cloud integration.

It shows how mobile technology can be used to solve an important social problem in a meaningful way. The application can have a positive impact on women's safety, student travel safety, elderly care, and emergency preparedness.

It helps improve user confidence by providing a sense of support and quick communication during critical moments. Although the current system has some limitations, it establishes a strong foundation for future improvement.

In future versions, AI-based voice detection can be added for better recognition accuracy and smart threat identification. Wearable integration with smartwatches or bands can provide faster and more discreet emergency activation.

The system can also be extended to connect directly with police stations, hospitals, or national emergency helplines. Live route prediction, safer path suggestion, and cloud dashboards can further enhance monitoring capabilities. Stronger data security and privacy controls can also improve trust and reliability.

Overall, Sanraksha is a socially relevant, technically sound, and impactful final year project with strong future scope in personal safety technology.